

PAPER_1625 — Earth Age = $T_{\oplus} = D + F \cdot D + F \cdot \Phi_{5/6} + F \cdot SSQ = 4.5403 \text{ Gyr}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Geophysics **Date:** June 18, 2026 **Status:** CLOSED — 0.007% closure

Observation

PAPER_1209CC_S611 (Geophysics Unified Proof Set) gives the formula:

$$T_{\oplus} = D + F \cdot D + F \cdot \Phi_{5/6} + F \cdot SSQ = 4.5403 \text{ Gyr}$$

Residual: **0.007%**.

UQFF Closed Identity

$$T_{\oplus} = D + F \cdot D + F \cdot \Phi_{5/6} + F \cdot SSQ = 4.5403 \text{ Gyr} \quad 0.007\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Tier Best — Earth Age 4.54 Gyr at 0.007%

The age of the Earth (= solar system) is derived from radiometric dating of CAI inclusions in meteorites (chondrules) and lunar zircons. Modern best estimate: **$4.54 \pm 0.05 \text{ Gyr}$** .

UQFF prediction:

$$\begin{aligned} T_{\oplus} &= D_{\text{phys}} + F_{\text{TRZ}} \cdot D_{\text{phys}} + F_{\text{TRZ}} \cdot \Phi_{5/6} + F_{\text{TRZ}} \cdot SSQ \\ &= 4 + 0.4 + 0.0833 + 0.057 \\ &= 4.5403 \text{ Gyr} \end{aligned}$$

Residual: **70 parts per million** from the geochronometric value. UQFF predicts **geological time itself** from the same integer primitives that govern particle masses, Standard Model fermion hierarchy, and atomic-scale EM constants.

This is one of the **tightest non-EXACT closures** in the entire UQFF catalog — the second-tightest tier match in PAPER_1209 series after PAPER_1556 (m_t at 0.005%) and PAPER_1518 ($MAD \eta_{\text{EM}}$ at <0.001%).

NOT REPLACEMENT

Empirical geophysics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209CC_S611
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "earth_age_4_54_gyr"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.